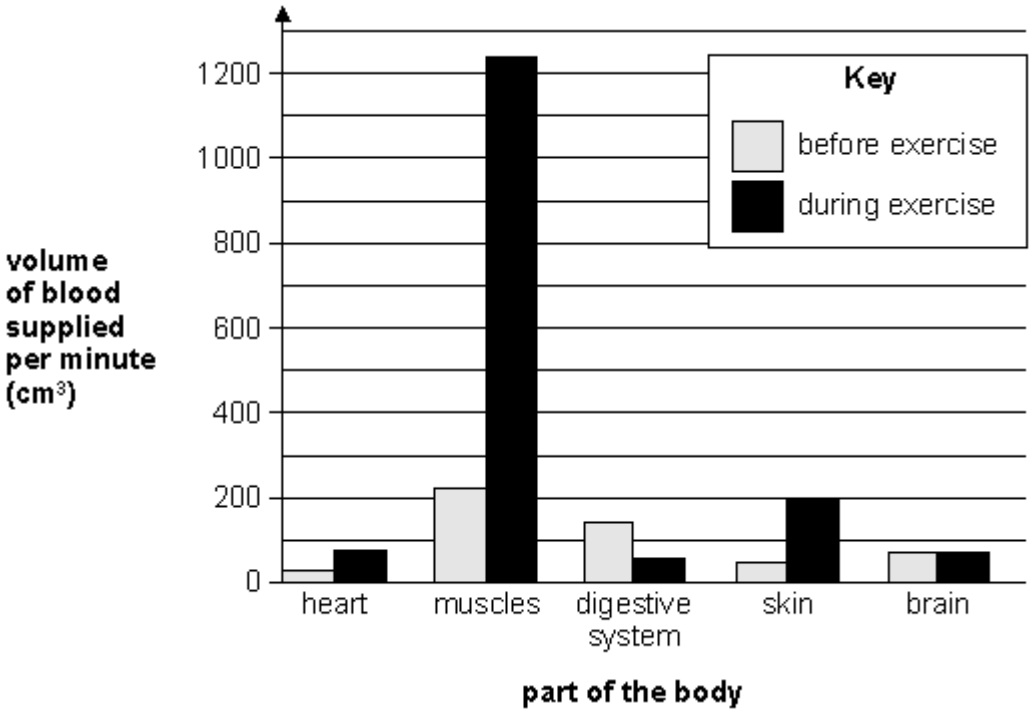
Q1. When people exercise, the volume of blood per minute needed to supply different parts of the body changes.

This is shown in the bar chart below.



(a) Explain why muscles need **more** blood during exercise. Give **three** reasons.

.....

.....

.....

.....

.....

3 marks

(b) Look at the bar chart.
Suggest why you should not go for a long run just after eating a meal.

.....

.....

1 mark

(c) Why is it important that the blood supply to the brain stays constant?

.....

.....

1 mark
maximum 5 marks

Mark schemes

Q1.

(a) any **three** from

- energy is released more quickly
  accept 'more energy is needed'
  'the muscles work harder' is insufficient

- cells respire more quickly
  accept 'respiration is quicker'
  *accept 'muscles **or** they respire more quickly'*
  'blood contains oxygen' is insufficient

- glucose is used up **or** supplied more quickly
  *accept 'more glucose is used **or** needed'*

- oxygen is used up **or** supplied more quickly
  *accept 'more oxygen is used **or** needed'*

- carbon dioxide **or** waste is produced **or** removed more quickly
  *accept 'more carbon dioxide is produced **or** removed'*

- water is removed more quickly
  *accept 'more water is produced **or** removed'*

- blood transports oxygen **or** glucose **or** carbon dioxide **or** waste
  *accept, for **two marks**, 'blood transports more oxygen' **or** 'blood carries carbon dioxide more quickly' as they cover the fourth **or** fifth and seventh marking points*

- heat is produced **or** removed more quickly
  *accept 'more heat is produced **or** removed'*

3 (L7)

(b) any **one** from

- less food is absorbed
  'less blood gets to the digestive system' is insufficient

- the digestive system slows down
  accept 'less food is digested'
  *accept 'it may cause a stitch **or** cramp'*
  accept 'less blood goes to the muscles than is needed'
  *do **not** accept 'no food can be digested **or** absorbed'*

1 (L7)

(c) any **one** from

- the brain **or** brain cells will die if there is less **or** no oxygen **or** glucose
  accept 'otherwise brain damage could occur'
  'the brain will die' is insufficient as it does not explain what the brain needs

- the person might faint **or** feel dizzy **or** lack concentration
- the brain needs a regular supply of oxygen **or** glucose

1 (L7)

[5]